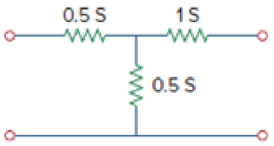
Chapter 19, Problem 37P

Problem

The input port of the circuit in Fig. 19.79 is connected to a 10-V dc voltage source while the output port is terminated by a 5-Ω resistor. Find the voltage across the 5-Ω resistor by using *h* parameters of the circuit. Confirm your result by using direct circuit analysis.

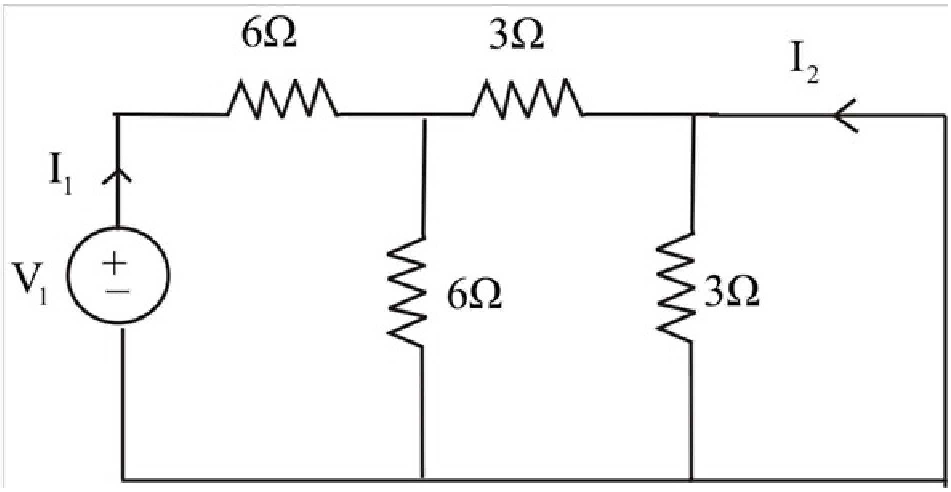
Figure 19.79



Step-by-step solution

Step 1 of 5

By making $V_2 = 0$

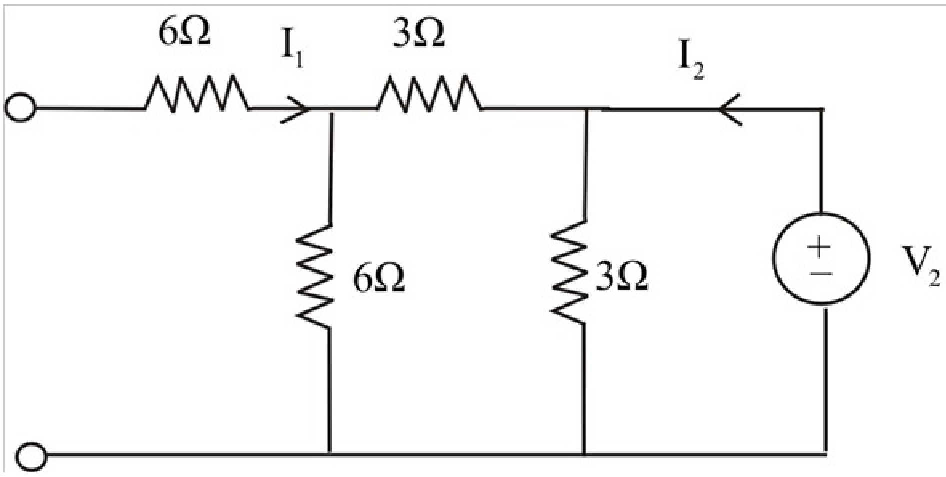


Step 2 of 5

$$h_{11} = \frac{V_1}{I_1} = \frac{V_1}{\frac{V_1}{(6\|3+6)}}$$

$$h_{21} = \frac{I_2}{I_1} = \frac{\frac{(I_1 \times 6)}{9}}{I_1} = \frac{-2}{3}$$

By making $I_1 = 0$ the circuit becomes



Step 3 of 5

$$h_{12} = \frac{V_1}{V_2} = \frac{\frac{\{[I_2 \times 3]\}}{9} \times 6}{I_2[9\|3]}$$

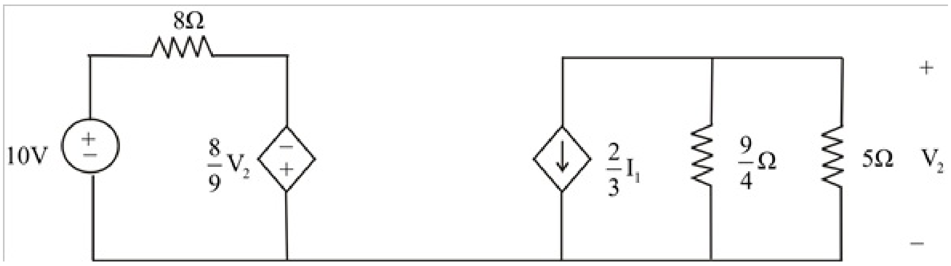
$$= \frac{2}{\frac{27}{12}} = \frac{8}{9}$$

$$h_{22} = \frac{I_2}{V_2} = \frac{I_2}{I_2(9\|3)} = \frac{8}{9}$$

$$h_{22} = \frac{I_2}{V_2} = \frac{I_2}{I_2(9\|3)} = \frac{12}{27} = \frac{4}{9}S$$

Step 4 of 5

Now the circuit becomes



Step 5 of 5

$$I_1 = \frac{10 - \frac{8}{9}V_2}{8}$$

$$V_2 = 5 \times \frac{\left[\frac{2}{3} I_1 \times \frac{9}{4}\right]}{\left[\frac{9}{4} + 5\right]}$$

$$V_2 = \frac{\frac{3}{2} I_1 \times 5}{\frac{29}{4}} + \frac{6 \times 5}{29} I_1 = \frac{30}{29} I_1$$

$$V_2 = \frac{6 \times 5}{9}\left[\frac{10 - \frac{8}{9}V_2}{8}\right] = \frac{-6 \times 5}{29 \times 8}\left[\frac{90 - 8V_2}{9}\right]$$

$$V_2 = \left[1 + \frac{6 \times 8 \times 5}{29 \times 8 \times 9}\right] = \frac{90 \times 6 \times 5}{29 \times 8 \times 9}$$

$$\boxed{V_2 = 1.159V}$$